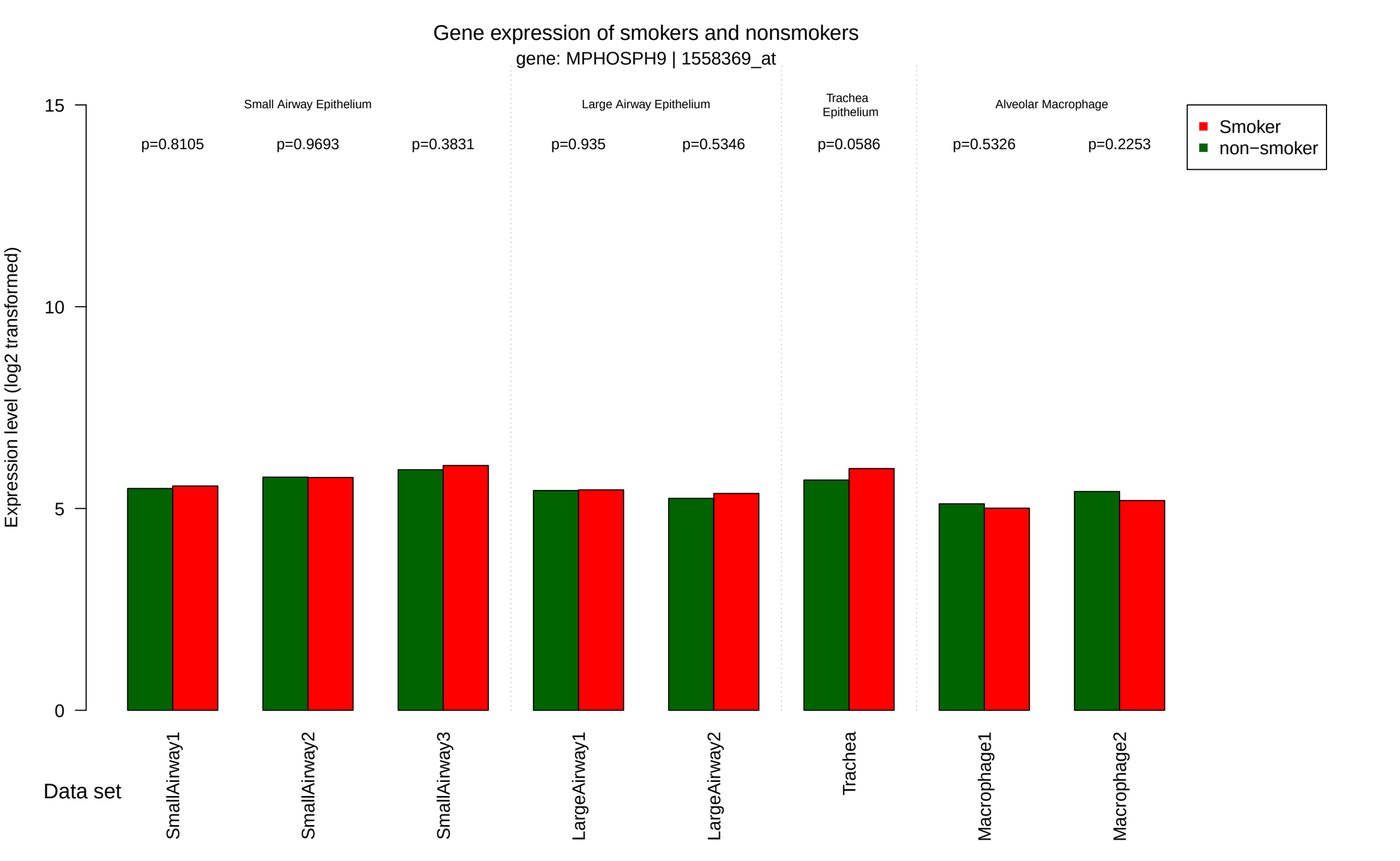
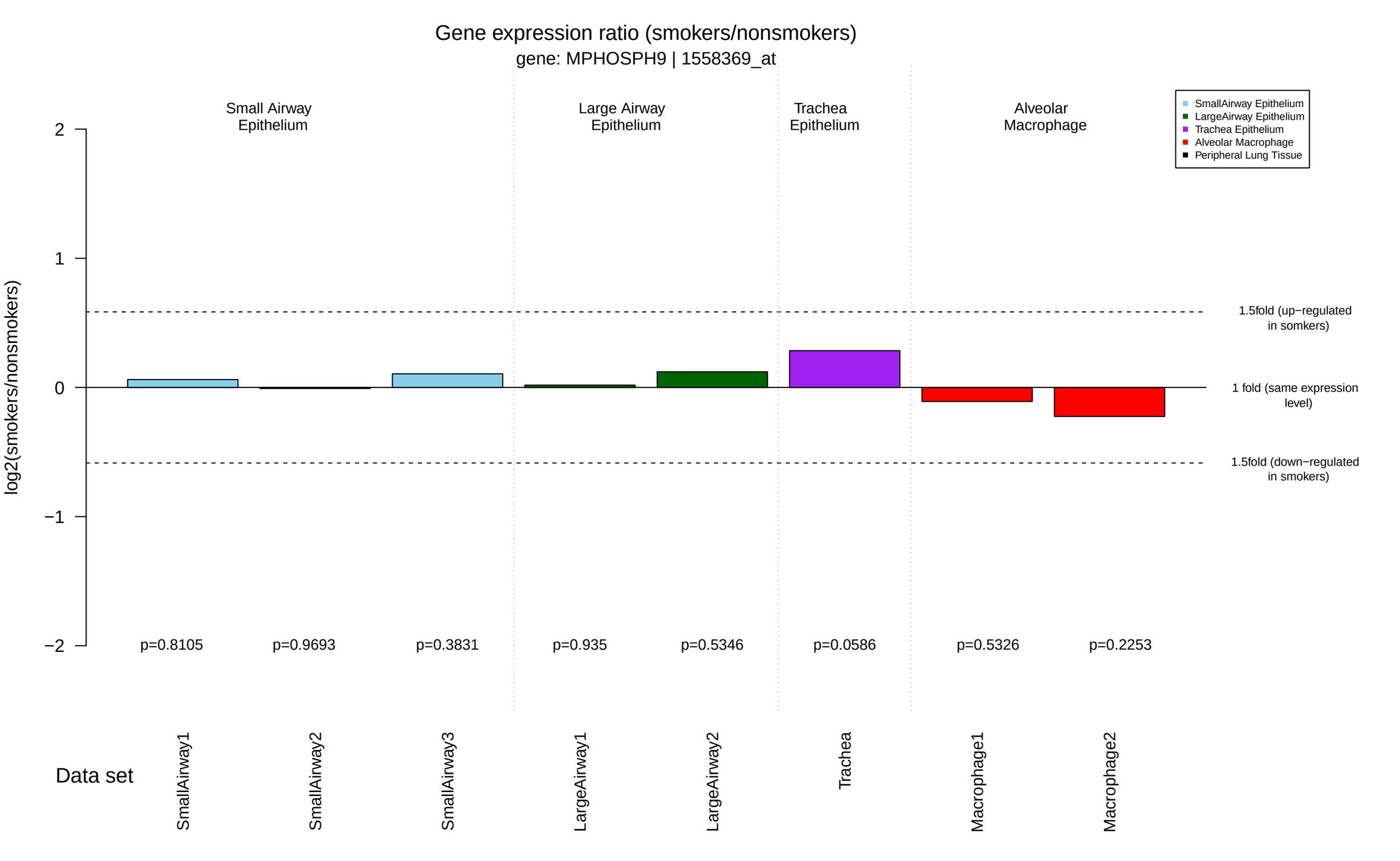
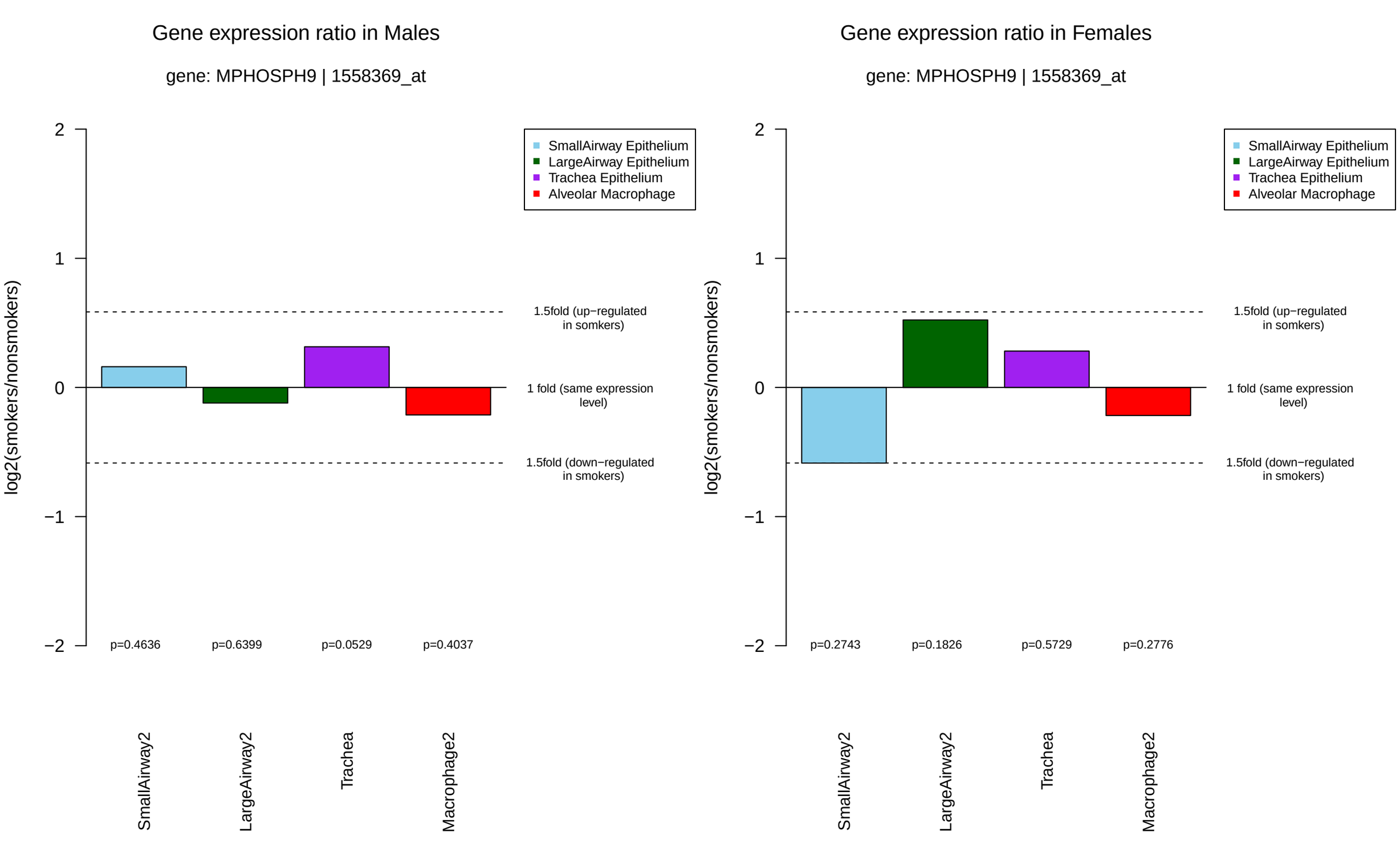
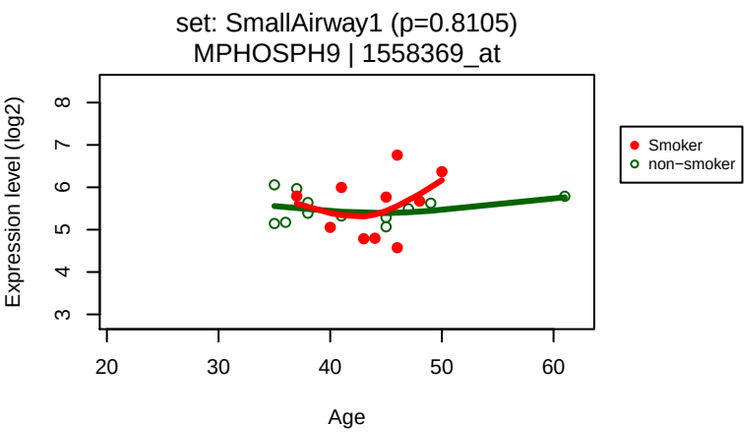
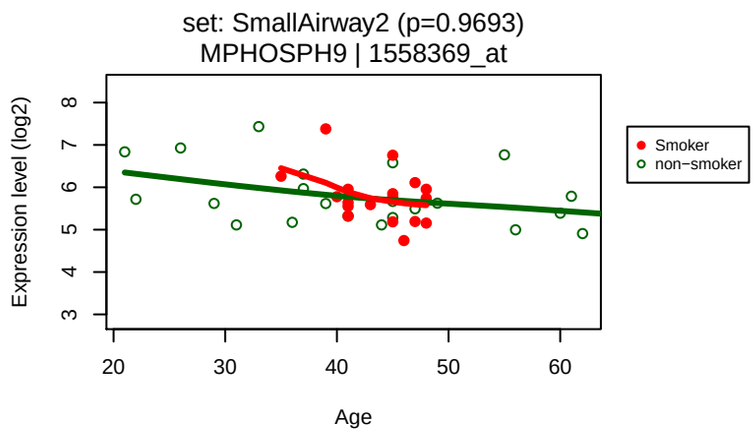
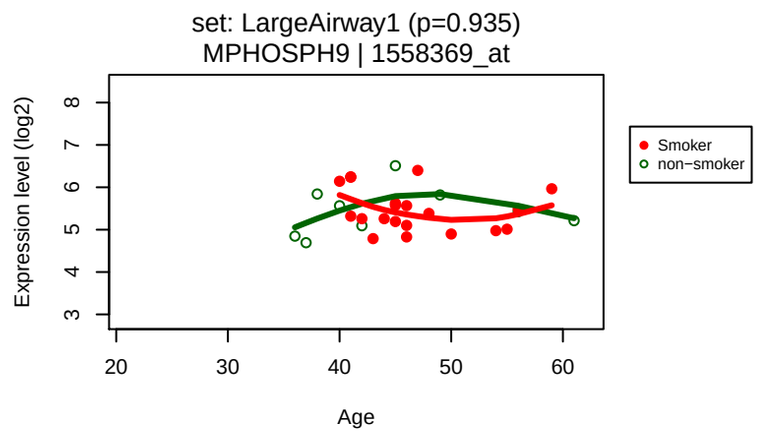

Gene expression of smokers and nonsmokers
gene: MPHOSPH9 | 1558369_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8105
p=0.9693
p=0.3831
p=0.935
p=0.5346
p=0.0586
p=0.5326
p=0.2253
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

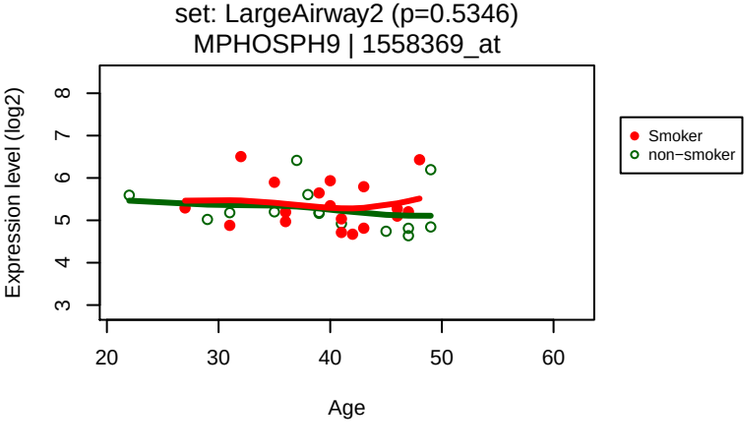

Gene expression ratio (smokers/nonsmokers)
gene: MPHOSPH9 | 1558369_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8105
p=0.9693
p=0.3831
p=0.935
p=0.5346
p=0.0586
p=0.5326
p=0.2253
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

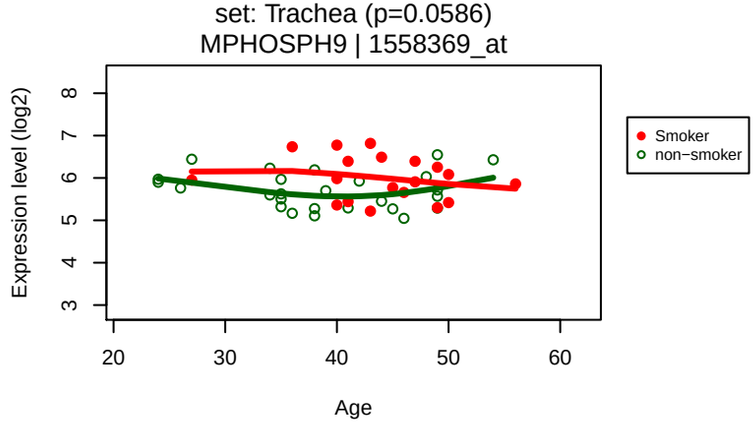

Gene expression ratio in Males
gene: MPHOSPH9 | 1558369_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4636
p=0.6399
p=0.0529
p=0.4037
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MPHOSPH9 | 1558369_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2743
p=0.1826
p=0.5729
p=0.2776
SmallAirway2
LargeAirway2
Trachea
Macrophage2

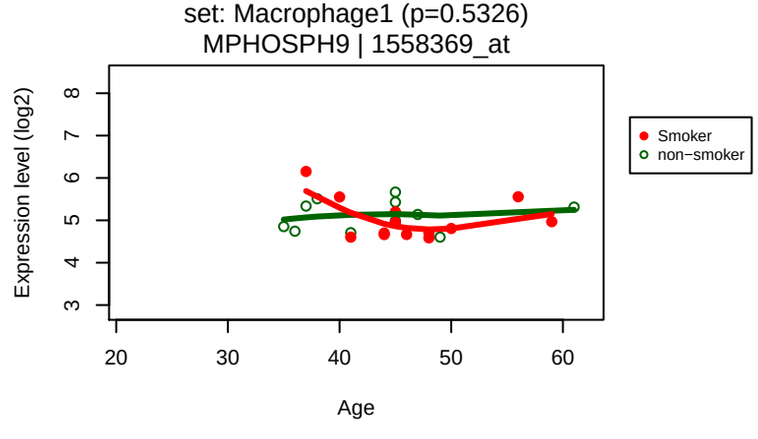

set: SmallAirway1 (p=0.8105)
MPHOSPH9 | 1558369_at
Expression level (log2)
Age
Smoker
non−smoker

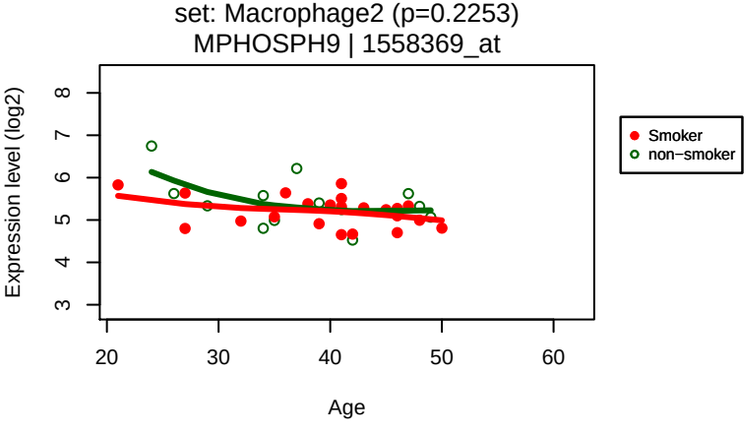

set: SmallAirway2 (p=0.9693)
MPHOSPH9 | 1558369_at
Expression level (log2)
Age
Smoker
non−smoker

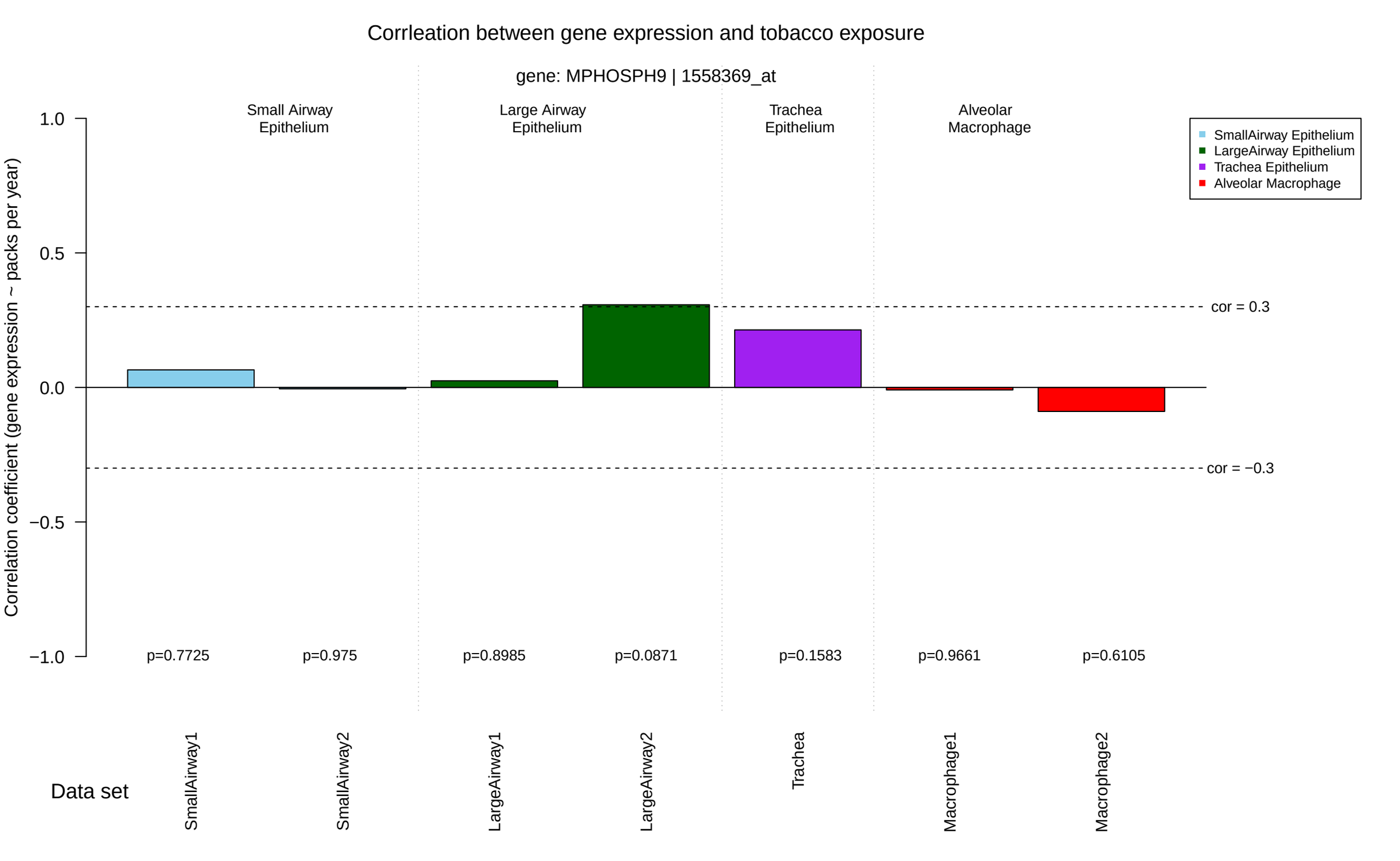

set: LargeAirway1 (p=0.935)
MPHOSPH9 | 1558369_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5346)
MPHOSPH9 | 1558369_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0586)
MPHOSPH9 | 1558369_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5326)
MPHOSPH9 | 1558369_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2253)
MPHOSPH9 | 1558369_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MPHOSPH9 | 1558369_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7725
p=0.975
p=0.8985
p=0.0871
p=0.1583
p=0.9661
p=0.6105
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2